Deep Reinforcement Learning Enabled Persistent Surveillance With Energy-Aware UAV-UGV Systems for Disaster Management Applications

Md Safwan Mondal, Subramanian Ramasamy, James D. Humann, James M. Dotterweich, and Pranav A. Bhounsule

***Abstract*—Integrating Unmanned Aerial Vehicles (UAVs) with Unmanned Ground Vehicles (UGVs) provides an effective solution for persistent surveillance in disaster management. UAVs excel at covering large areas rapidly, but their range is limited by battery capacity. UGVs, though slower, can carry larger batteries for extended missions. By using UGVs as mobile recharging stations, UAVs can extend mission duration through periodic refueling, leveraging the complementary strengths of both systems. To optimize this energy-aware UAV-UGV cooperative routing problem, we propose a planning framework that determines optimal routes and recharging points between a UAV and a UGV. Our solution employs a deep reinforcement learning (DRL) framework built on an encoder-decoder transformer architecture with multi-head attention mechanisms. This architecture enables the model to sequentially select actions for visiting mission points and coordinating recharging rendezvous between the UAV and UGV. The DRL model is trained to minimize the *age periods* (the time gap between consecutive visits) of mission points, ensuring effective surveillance. We evaluate the framework across various problem sizes and distributions, comparing its performance against heuristic methods and an existing learning-based model. Results show that our approach consistently outperforms these baselines in both solution quality and runtime. Additionally, we demonstrate the DRL policy's applicability in a real-world disaster scenario as a case study and explore its potential for online mission planning to handle dynamic changes. Adapting the DRL policy for priority-driven surveillance highlights the model's generalizability for real-time disaster response.**

***Index Terms*—Disaster management, deep reinforcement learning, UAV, UGV, surveillance, cooperative routing, multi-agent systems, combinatorial optimization.**

Received 25 October 2024; revised 28 September 2025; accepted 2 February 2026. This work was supported by ARO under Grant W911NF-14-S-003. *(Corresponding author: Md Safwan Mondal.)*

Md Safwan Mondal, Subramanian Ramasamy, and Pranav A. Bhounsule are with the Department of Mechanical and Industrial Engineering, University of Illinois Chicago, Chicago, IL 60607 USA (e-mail: mmonda4@uic.edu; sramas21@uic.edu; pranav@uic.edu).

James D. Humann is with the DEVCOM Army Research Laboratory, Minneapolis, MN 55455 USA (e-mail: james.d.humann.civ@army.mil).

James M. Dotterweich is with the DEVCOM Army Research Laboratory, Aberdeen Proving Grounds, Aberdeen, MD 21005 USA (e-mail: james.m.dotterweich.civ@army.mil).

More details are available on our website: https://sites.google.com/view/ecucps, provided by the authors.

This article has supplementary downloadable material available at https://doi.org/10.1109/TIV.2026.3696943, provided by the authors.

Digital Object Identifier 10.1109/TIV.2026.3696943

I. Introduction

UNMANNED Aerial Vehicles (UAVs) have emerged as transformative agents in modern disaster management systems due to their speed, mobility, cost-efficiency, and ability to navigate challenging environments [1], [2]. With increasing frequency of natural disasters worldwide, the need for rapid and effective disaster response is crucial to save lives and prevent substantial losses. This necessitates continuous monitoring and data collection at disaster sites, where UAV-based intelligence, surveillance, and reconnaissance (ISR) missions have become integral to contemporary disaster management strategies [3], [4]. UAVs provide vital real-time data and continuously assess scenarios to support both planning and operational phases of disaster response. Notable deployments include damage assessment during the 2015 Nepal earthquake in areas inaccessible to rescue teams [5], flood monitoring and rescue operations during the 2018 Kerala floods [6], and COVID-19 pandemic surveillance for monitoring gatherings and delivering medical supplies in Indian cities like Delhi and Bengaluru [7].

Despite their advantages, UAVs face significant limitations due to restricted battery life, hindering continuous operation during prolonged missions. While UAVs can be recharged at fixed depots, the stationary nature of these refueling points limits operational range. Consequently, Unmanned Ground Vehicles (UGVs) are employed as mobile recharging bases, providing ground support and extending UAV operational capabilities along designated roadways [8], [9], [10]. This UAV-UGV synergy enhances persistent surveillance efficiency and comprehensive disaster management coverage. Our paper explores a UAV-based persistent surveillance problem integrated with a UGV to overcome endurance limitations and maximize operational impact in disaster scenarios. In this energy-constrained UAV-UGV cooperative persistent surveillance (**ECUCPS**) system, an energy-limited UAV departs from a designated depot to visit mission points and collect information (see Fig. 1). For sustained operations, the UAV is periodically recharged by a UGV that also departs from the same depot and travels along the road network to visit mission points located on roads. The objective is to minimize time gaps between consecutive visits to mission points by maximizing visit frequency of either vehicle. This ECUCPS system is crucial for disaster management scenarios where continuous, up-to-date information collection from

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

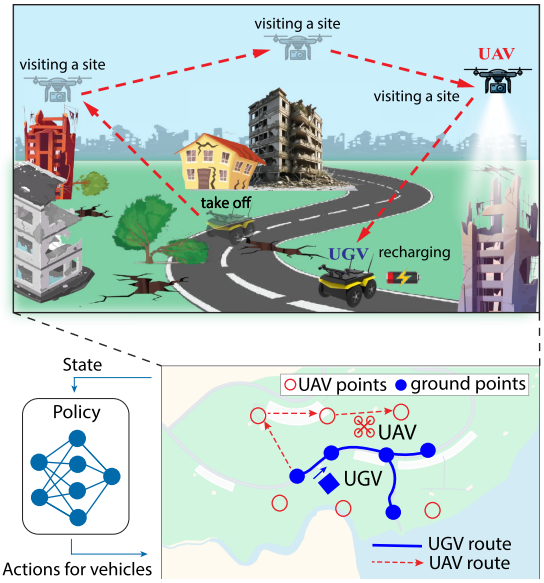


Fig. 1. Illustration of cooperation between an energy-limited UAV and a UGV for monitoring disaster-affected regions. The UAV conducts continuous surveillance and refuels through the UGV. The proposed DRL policy determines mission point visits and coordinates UAV-UGV recharging.

disaster sites is essential, ensuring critical mission points are frequently visited to provide reliable real-time data supporting effective disaster response efforts.

II. Related Works

We review the prior research on similar routing problems in two areas: UAV-based routing for persistent surveillance and recent advancements in learning-based approaches for routing problems.

A. UAV Based Persistent Surveillance Problem

UAV routing has received significant attention in disaster management due to its ability to address complex logistical and surveillance challenges. Classical formulations such as the Traveling Salesman Problem with Drones (TSP-D) and the Vehicle Routing Problem with Drones (VRP-D) have been widely used to develop efficient UAV routing solutions [11], [12], [13], [14]. Scherer et al. [15] studied energy and communication constraints for multi-UAV surveillance and analyzed the effect of base station placement on mission performance. Angun and Dundar [16] formulated UAV surveillance as an energy-capacitated location-routing problem for disaster risk mitigation. Calamoneri et al. [17] proposed a multi-depot, multi-trip VRP variant to enable repeated UAV visits to disaster sites. Nigam et al. [18] addressed persistent surveillance by decomposing 2D areas into sensor-footprint-based grids and assigning UAVs for repeated monitoring. Stump and Michael [19] modeled surveillance over discrete targets using VRP with Time Windows and a receding-horizon strategy. Despite these advancements, UAV fuel limitations remain relatively underexplored. Hari et al. [20] introduced models restricting UAV visits to reflect fuel constraints, determining minimal maximum revisit durations among target nodes. Traditional setups assume fixed service stations for UAV recharging, but recent studies suggest employing mobile UGVs as dynamic recharging units, known as the UAV-UGV cooperative routing problem. This integration of heterogeneous vehicles complicates the problem, as being NP-hard, it becomes intractable as problem size grows [21], making exact MILP solutions impractical due to scalability issues. Consequently, heuristic methods such as genetic algorithms, tabu search, and ant colony optimization have been devised to produce high-quality approximate solutions within acceptable timeframes [22], [23], [24], though these are typically tailored to specific problem types, limiting their generalizability. Seyedi et al. [8], [25] developed heuristics organizing uniform UAV-UGV teams to cyclically patrol optimally partitioned surveillance areas, though this strategy may not suit discrete mission point scenarios due to spatial partitioning reliance. In heterogeneous vehicle cooperative routing, multi-level optimization approaches like "UGV first, UAV second" simplify complexity by breaking problems into manageable subproblems. Maini et al. [10], [26] implemented bi-level strategies, initially identifying potential rendezvous points via minimum set cover problems, then formulating MILP for optimal UAV-UGV routes. Nigam et al. [27] proposed value-sum strategies directing UAVs toward areas with highest cumulative value or target-based tactics focusing on cells with greatest singular value. Similarly, Chour et al. [28] tackled UAV-UGV rendezvous through multi-level coordination algorithms, solving reward-based TSP using receding horizon approaches for UAV routes. However, this method suffers from the *horizon effect* [29], where locally optimal routes may prove suboptimal over complete mission durations.

B. DRL for UAV-UGV Cooperative Routing

In recent years, learning-based methodologies have emerged as a promising alternative for addressing routing problems and other combinatorial optimization challenges [30], [31], [32], [33]. Deep reinforcement learning (DRL) approaches have been employed for TSP and VRP due to their ability to autonomously discover optimal routing strategies, handle complex constraints, and adapt to diverse scenarios through continuous learning. Most DRL frameworks involve end-to-end encoding and decoding processes where the encoder extracts node features from inputs, while the decoder sequentially selects node sequences, training the neural model through reinforcement learning to preferentially output high-profit nodes at each time step. Vinyals et al. [34] pioneered the Pointer Network (PN), enhancing RNNs with attention mechanisms [35] and training them in a supervised manner for TSP. However, supervised learning methods are constrained by the need for extensive and costly labels generated by pre-solving routing problems. Addressing these limitations, Kool et al. [36] introduced a Transformer-based encoder-decoder architecture that outperformed traditional heuristic methods across various routing problems. Similarly, Li et al. [37] utilized DRL with attention mechanisms for heterogeneous capacitated VRP, while Wu et al. [38] explored truck-and-drone delivery using encoder-decoder framework. In

parallel, GNN-based approaches have also been applied to routing and combinatorial optimization [39], [40]. Joshi et al. [41] proposed a graph convolutional network for TSP that constructs efficient graph representations and generates tours non-autoregressively, achieving good quality and inference speed. Fan et al. [42] employed multi-head attention with DRL for energy-constrained UAV routing, and Bana et al. [43] applied DRL to persistent surveillance for fuel-limited UAVs, though both relied on fixed recharging stations. While DRL has successfully addressed various vehicle routing problems, UAV-UGV collaborative routing for persistent surveillance remains relatively unexplored. Unlike traditional routing where previously visited points are excluded, persistent surveillance requires continuous revisitation of mission points, creating constant action spaces and added complexity. Additionally, incorporating UGVs as mobile recharging stations expands potential recharging locations, leading to larger and more dynamic solution spaces compared to fixed stations. Consequently, UAV-UGV cooperative surveillance becomes computationally more challenging, requiring advanced optimization techniques. In cooperative routing, determining optimal mission point sequences and strategically planning UAV-UGV recharging instances for seamless coordination is essential, as demonstrated by Ramasamy et al. [44]. Building on our prior work [45] on DRL-based cooperative routing framed as a TSP problem, this paper proposes a deep reinforcement learning framework based on an encoder-decoder transformer architecture and policy gradient method to determine coordinated routing between an energy-constrained UAV and a UGV acting as a mobile recharging station. The framework has been tested across different problem sizes and distributions, demonstrating significant adaptability and generalizability for UAV-UGV cooperative routing in persistent surveillance scenarios. The key contributions of this work include:

1) *Problem Modeling:* We first model the ECUCPS problem using a multi-level optimization strategy for approximate heuristic solutions and then as a Markov Decision Process (MDP) for solving it through reinforcement learning (RL).
2) *Framework Evaluation:* We introduce an encoder-decoder-based transformer architecture with attention layers and dynamic node embeddings to tackle the ECUCPS problem. The framework's generalization capabilities are demonstrated through systematic evaluations across various problem sizes and distributions, and its applicability is further validated in a simulation case study of a real-world disaster scenario.
3) *Comparison with Existing Methods:* We compare our method against heuristic-based approaches for persistent surveillance routing and a learning-based framework as baselines. Our approach demonstrates superior solution quality, highlighting its effectiveness and reliability over the baselines.
4) *Online and Priority-Driven Route Planning:* We investigate the framework's utility for online route planning in response to dynamically appearing mission points and also extend its application for prioritized visits to critical mission points in weighted persistent surveillance scenarios.

The remainder of this paper is structured as follows: Section III presents the problem statement and Markov Decision Process (MDP) formulation. Section IV describes problem modeling and a heuristics approach for persistent surveillance. Section V details the proposed method and policy network. Section VI presents and analyzes our experimental results. Finally, Section VII concludes the paper and discusses future work.

III. Problem Overview

A. Problem Statement

In this study, we examine the UAV-UGV collaborative routing problem for disaster management, focusing on persistent surveillance that continuously monitors an area to capture real-time information. We consider a system composed of a UAV U_a and a UGV U_g tasked with visiting n mission points $\mathcal{M} = \{m_1, m_2, \ldots, m_n\}$ spread over a region (see Fig. 3(a)). There are two types of mission points: ground points $\mathcal{M}_g$ located within road network G that can be surveyed by either UAV flyover or UGV road visit, and UAV points $\mathcal{M}_a$ situated outside the road network, accessible only to the UAV. Thus, $\mathcal{M} = \mathcal{M}_g \cup \mathcal{M}_a$.

Definition 1: The *age period* of any mission point is the time gap between the current time and the last visit to that mission point:

$$\mathfrak{a}^k = t - t^k_{\text{last}} \tag{1}$$

where $\mathfrak{a}^k$ is the *age period* for mission point $m_k \in \mathcal{M}$ at time t and t^k_{last} is the last visit time.

Definition 2: To reinforce persistent visits and ensure recent data collection from disaster sites, we define the score metric as:

$$\mathbb{S} = \frac{1}{\alpha}\sum_{k=1}^{n}\sum_{q=1}^{p}\left(t^k_q - t^k_{q-1}\right)^2 = \frac{1}{\alpha}\sum_{k=1}^{n}\sum_{q=1}^{p}(\mathfrak{a}^k_q)^2 \tag{2}$$

Here, $t^k_0, t^k_1, \ldots, t^k_p$ are the time instances when mission point m_k is visited during the mission period in ascending order. The parameter $\alpha = T_m^2$ scales down the score metric value where T_m is the mission period duration. The score metric penalizes longer intervals between consecutive visits through its square function design.

The UAV and UGV have heterogeneous characteristics: the UAV has limited battery capacity F^a but faster velocity v_a, while the UGV moves at slower pace v_g and operates exclusively on road network G. The UGV acts as a mobile recharging station, refueling the UAV at any ground point where the UAV spends fixed recharging time T_R before resuming flight. Both vehicles operate continuously to visit mission points and rendezvous for recharging until the mission period concludes.

Problem 1: The objective is to devise a strategy for collaborative operation between the UGV and the UAV to ensure all mission points are consistently visited during the mission, thereby minimizing the score metric at the end of the mission planning horizon. This challenge is multifaceted, requiring the optimization of the UGV and UAV route sorties to visit mission points and scheduling their recharging instances to synchronize them in both time and location.

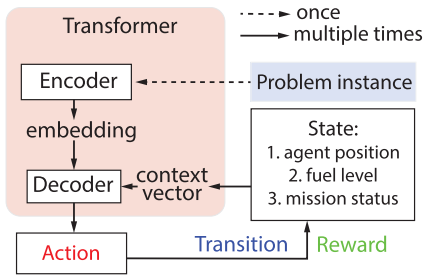


Fig. 2. MDP representation for the UAV-UGV cooperative persistent surveillance problem utilizing a Transformer architecture.

B. Markov Decision Process (MDP) Formulation

The problem can be modeled as a sequential decision-making system where the UAV sequentially selects the mission points to visit or recharge locations for rendezvous with the UGV (see Fig. 2). This system can be formulated as a Markov Decision Process (MDP), with its components defined by the tuple $< \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T} >$, as follows:

1) *State Space* ($\mathcal{S}$)*:* At any decision making step, the state of the environment $s_t \in \mathcal{S}$ is defined as, $s_t = (p_t, f_t,\ q_t)$. Here, $p_t = \{x_t, y_t\}$ represents the current position of the UAV and f_t indicates its fuel level. Additionally, $q_t = \{x^i, y^i, \mathfrak{a}_t^i\}$ highlights the coordinates and *age period* status of the mission points $m_i \in \mathcal{M}$. The UGV's state is not represented explicitly, since its trajectory is fully determined by the UAV's recharge decisions, once the UAV selects a recharge action, the UGV moves to the corresponding rendezvous point.

2) *Action Space* ($\mathcal{A}$)*:* The UAV agent can perform two types of actions, $a_t \in \mathcal{A}$: *visiting* (on both UAV points $\mathcal{M}_a$ and ground points $\mathcal{M}_g$) or *recharging* (only on ground points $\mathcal{M}_g$). Since $\mathcal{M}_g$ is used for both visiting and recharging, the action space is defined as, $\mathcal{A} = \{\mathcal{M}_g(\text{for recharging}) + \mathcal{M}_g(\text{for visiting}) + \mathcal{M}_a(\text{for visiting})\}$. Infeasible actions based on the current state s_t at time t are masked out from the action space (details in subsection V-A2).

3) *Return* ($\mathcal{R}$)*:* To align with the objective of minimizing the total score metric, at each decision-making step t, the reward r_t is calculated as the square of the *age period*, $r_t = (\mathfrak{a}_t^i)^2$, where $\mathfrak{a}_t^i$ represents the *age period* of the mission point m_i selected as the action a_t at step t. At the end of the mission planning horizon the total return for the mission is computed as the sum of all such rewards over the entire mission planning horizon: $\mathcal{R} = \sum_{t=0}^{T_m} r_t$. Finally, to replicate the original score metric function the total return is divided by the square of the total mission period time T_m as $\mathcal{R} = \frac{1}{T_m^2} \sum_{t=0}^{T_m} r_t$.

4) *Transition* ($\mathcal{T}$)*:* The transition function updates the current state s_t to the next state $s_{t+1} = (p_{t+1}, f_{t+1}, q_{t+1})$ based on the taken action a_t at time t. The UAV's position is updated to the chosen task location, $p_{t+1} = (x_{t+1}, y_{t+1}) \equiv a_t$. Mission time is incremented by the travel time between nodes and for recharge actions, by the recharging service time: $t = t_{ij} + T_R$, where $T_R = 0$ for visiting actions. For recharge actions, the cooperative nature of the problem is captured through waiting time. If the UAV reaches the rendezvous point earlier than the UGV, it must remain idle until the UGV arrives via the road network. This synchronization reflects the dependence of UAV operations on UGV mobility, as the slower ground vehicle constrains when and where recharging can occur. The UAV fuel level is updated as $f_{t+1} = f_t - f_{ij}$ for visiting tasks, or reset to $f_{t+1} = F^a$ for recharging. The *age period* status of all mission points is updated as $\mathfrak{a}_{t+1}^i = t - t_{last}^i$, for all mission points $m_i \in \mathcal{M}$, and for the selected action $m_j \equiv a_t$, the last-visit time is set to $t_{\text{last}}^j = t$.

IV. Problem Modelling

In this section, we model the persistent surveillance problem using a bilevel optimization strategy as a foundational approximation. Following this, we introduce heuristics based baseline methods using a standard solver for comparison. Finally, we present our Deep Reinforcement Learning (DRL) approach as a more adaptive solution to the problem.

A. Bilevel Optimization Framework

The bilevel optimization framework provides a structured approach for solving the UAV-UGV cooperative routing problem in persistent surveillance. These multi-echelon optimization strategies are commonly employed in heterogeneous vehicle routing problems, typically using sequencing strategies such as "UGV first, UAV second", "UAV first, UGV second," or other vehicle-prioritization schemas [10], [22], [26]. The routing problem is tackled in stages: in the first stage (outer level), the route for one vehicle type is planned based on mission parameters and environmental conditions, serving as a reference for the subsequent stage. In the second stage (inner level), the route for the second vehicle is optimized, considering interactions with the previously planned route and adjusting for task allocation, timing, and spatial coordination. This hierarchical approach simplifies the complex cooperative routing problem by breaking it into manageable subproblems, each tackled using heuristic methods or optimization techniques suited to the specific vehicle characteristics. For our persistent surveillance problem, we utilize the "UGV first, UAV second" optimization strategy. In the first stage, we fix the UGV route, considering its slower speed, road network constraints, and role as a mobile recharging station. In the second stage, we construct the UAV route based on the UGV's route, optimizing for battery constraints, recharging coordination, and continuous mission point revisitation for real-time data collection.

1) Outer Level. UGV Routing: First, the UGV's traversal direction from the starting depot is determined by solving a Traveling Salesman Problem (TSP) on the road network G, connecting the depot with the extreme end nodes of each branch. The resulting TSP solution specifies the optimal direction for the UGV, which it follows during persistent surveillance to cover the entire road network in minimum time, as illustrated in Fig. 3(a). Using this travel direction, we identify potential refueling locations where the UAV and UGV can rendezvous. Let x_0 be the UGV's current position and $\vec{X}_g$ be its travel direction. Locations on the road network along $\vec{X}_g$ that the UGV can access within the UAV's maximum flight duration T_f at speed v_g are considered potential refueling stops. The set of

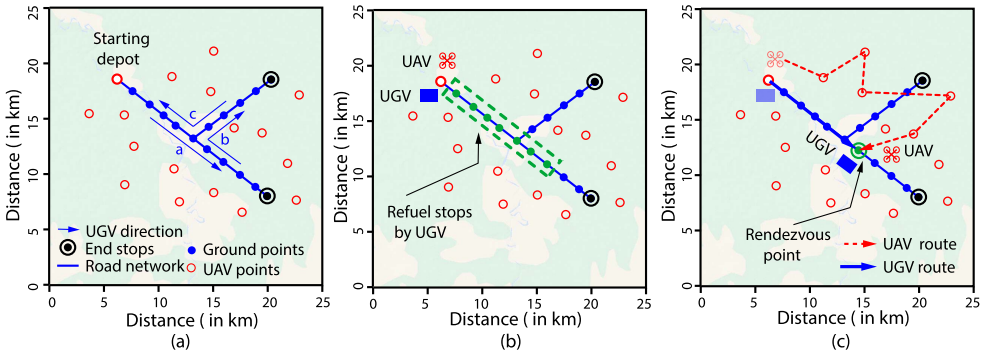


Fig. 3. Bilevel optimization process: a) Initial scenario with UAV and ground points, along with the UGV's movement direction along the road network from the starting depot, derived from the TSP solution. b) Refueling stops offered by the UGV during O-EVRPTW planning for the UAV. c) Recharging meetups between the UAV and UGV, with their respective route sorties, resulting from the O-EVRPTW solution.

possible refuel stops X^r is computed as:

$$X^r = \left[\left\{ x_i \mid \frac{||x_i - x_0||}{v_g} \leqslant T_f \right\}, \forall\, x_i \in G_{\vec{X_g}} \right] \tag{3}$$

Once potential refuel stops are identified (Fig. 3(b)), time windows are calculated. Since the UAV can only land after the UGV reaches a refueling stop, the lower time limits are:

$$t_l^r = \left[\left\{ \frac{||x_i^r - x_0||}{v_g} \right\}, \forall\, x_i^r \in X^r \right] \tag{4}$$

These time windows synchronize UAV-UGV rendezvous at the chosen refuel stop, determined during UAV routing. After identifying potential stops and time windows, this information is sent to the inner level for UAV routing, which provides rendezvous location and timing details. The UGV is then connected from its current position to the chosen refuel location to obtain spatial route information (Fig. 3(c)). Assuming constant UGV velocity, temporal route information is also derived. If the UGV arrives before the UAV, it waits at the refuel stop. This process repeats iteratively for each UAV route sortie planning during the persistent surveillance mission period.

2) Inner Level. UAV Routing: At the second stage, we obtain available potential refueling stops X^r and their time windows t_l^r from the previous UGV routing level. We model the UAV routing problem as an open-ended, energy-constrained vehicle routing problem with time window constraints (O-EVRPTW), aiming to minimize the *age period* of mission points. The O-EVRPTW formulation uses graph theory where mission nodes $\mathcal{M}$ and refueling stops X^r serve as vertices $V = \mathcal{M} \cup X^r$, with edges $E = \{(i,j) \parallel i,j \in V, i \neq j\}$ and arc costs t_{ij} (travel time). Decision variable x_{ij} indicates vehicle movement from vertex i to j, while binary variable y_i determines if node i is visited. The UAV starts at point S, visits mission points, and stops to recharge at a refueling stop X^r subject to time window constraints. The objective function, as defined in (5), is to minimize the cumulative travel duration while dropping the least number of mission points and prioritizing the mission points with higher *age periods* (not visited in recent times). This is achieved by imposing a penalty proportional to the square of the current *age period* of the dropped mission point, $P = (\mathfrak{a}^i)^2$. Eq. 6 ensures that only one refuel stop is chosen where the UAV ends its route for recharging. Energy constraints (7)–(9) ensure the UAV doesn't run out of fuel and follows consumption rates. Time-window constraint (10) mandates the UAV reaches the UGV only after it arrives at the refueling stops. Cumulative arrival time at node j equals time at node i plus travel time t_{ij} (11). Miller-Tucker-Zemlin formulation with constants L_1 and L_2 prevents sub-tours. Other VRP constraints are detailed in our previous works [46], [47].

Objective:

$$\min \quad \sum_i \sum_j t_{ij} x_{ij} + P \sum_i (1 - y_i) \quad \forall i,j \in V \tag{5}$$

Major constraints:

$$\sum_i y_i = 1 \quad \forall i \in X^r \tag{6}$$

$$f_i^a = F^a, \quad i \in X^r \tag{7}$$

$$0 \leq f_i^a \leq F^a, \quad \forall i \in V \setminus \{S, X^r\} \tag{8}$$

$$f_j^a \leq f_i^a - (f_{ij} x_{ij}) + L_1 (1 - x_{ij}), \quad \forall i,j \in V \setminus \{S, X^r\} \tag{9}$$

$$t_i \geq t_{l,i}^r \quad \forall i \in X^r \tag{10}$$

$$t_j \geq t_i + (t_{ij} x_{ij}) - L_2 (1 - x_{ij}), \quad \forall i,j \in V \tag{11}$$

The UAV route sortie is calculated by solving the O-EVRPTW formulation. The endpoint coordinates and time define rendezvous location $x^{\mathcal{R}}$ and time $x^{\mathcal{R}}$ (Fig. 3(c)), passed back to finalize the UGV's sortie. At rendezvous, the UAV spends fixed recharging time T_R before continuing.

In persistent surveillance missions, we plan UGV and UAV routes in receding horizon fashion, where each cycle corresponds to the interval between consecutive recharging rendezvous. After each cycle, *age periods* are updated based on planned routes. Once UAV completes recharging, we proceed to the next planning phase for subsequent sorties, repeating iteratively until the mission planning horizon ends.

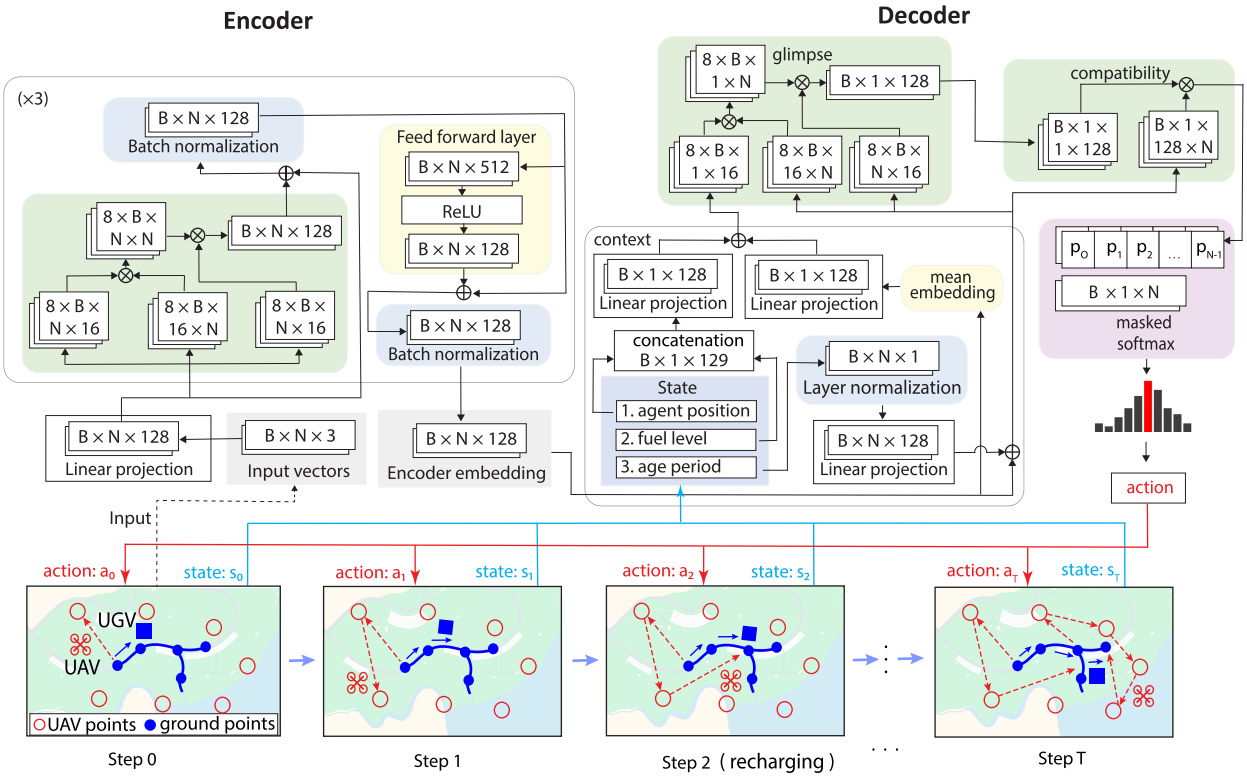


Fig. 4. The architecture of the proposed Transformer network consists of an encoder with three attention layers that create input embeddings from raw data while the decoder generates a context vector based on the current state. The network uses both input embeddings and the context vector, which are processed through multi-head and single-head attention layers to determine the next action, sequentially forming the cooperative route for persistent surveillance, as shown.

B. Baseline Heuristics

We establish a non-learning baseline for solving our ECUCPS problem using the bilevel optimization strategy outlined earlier. Specifically, we leverage the Google OR-Tools^{TM} CP-SAT solver [48], which utilizes constraint programming (CP) to solve the O-EVRPTW. To enhance the solver's performance and avoid local optima, we implement three metaheuristics within the OR-Tools framework: 1) **Guided Local Search (GLS)**, 2) **Tabu Search (TS)**, and 3) **Simulated Annealing (SA)**. These heuristics provide approximate suboptimal solutions within significantly lesser runtime compared to traditional MILP approaches, as demonstrated in similar cooperative routing problems [49]. This baseline approach serves as a benchmark for comparison with our proposed DRL-based methodology.

V. Reinforcement Learning Framework

This section presents a Deep Reinforcement Learning (DRL) methodology for solving the UAV–UGV cooperative persistent surveillance problem. We use an encoder–decoder transformer network trained with an RL algorithm to learn the UAV's routing policy π_θ, where θ are trainable parameters.

In this framework, the UAV acts as the primary decision-maker, visiting mission points and coordinating with the UGV, which serves as a mobile charging station. At each timestep t, the policy π_θ selects an action a_t for the UAV based on the scenario state s_t, until the terminal state s_T is reached. Although the policy is defined only for the UAV, cooperation arises when a recharge action is chosen, the rendezvous location is determined by the policy, and the UGV's route is constructed by linking successive rendezvous points. The resulting policy output is a persistent surveillance route $\mathcal{T}$, consisting of mission visits and recharge events, represented by the joint probability distribution:

$$\mathbb{P}(\mathcal{T};\theta) = \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t)\mathbb{P}(s_{t+1}|s_t,a_t) \tag{12}$$

Here, T is the timesteps till mission termination and $\mathbb{P}(s_{t+1}|s_t,a_t) = 1$, as we have chosen deterministic state transition.

A. Encoder-Decoder Transformer Architecture

For the routing policy π_θ, we employ an encoder-decoder transformer architecture [50], which excels in sequential data processing and parallel operations [51], [52], [53]. As shown in Fig. 4, we adapt this architecture for persistent surveillance: the encoder transforms mission point coordinates into high-dimensional embeddings for enhanced feature extraction, while the decoder uses these embeddings to make routing decisions based on contextual information from the current scenario state. This leverages the transformer's capacity to process complex sequential data, enabling effective learning and optimization of the UAV routing policy in our UAV-UGV cooperative framework.

1) Encoder: In the encoder, we have implemented multi-head attention (MHA) mechanism to achieve a higher-dimensional representation of the raw features in the problem

instance. The encoder takes the 3D vector representation of mission points as input, denoted by $X = (o_i = \{x_i, y_i, b_i\}, \forall\, m_i \in \mathcal{A})$, where (x_i, y_i) represents normalized coordinates and b_i is a binary variable that indicates if mission point m_i is eligible for recharging (ground points). Initially, each node input is linearly projected into a node embedding $h_i^0 = W^0 o_i + b^0$ with a dimension of $d_h = 128$, where W^0, b^0 are trainable parameters for the linear projection. This initial embedding h_i^0 is then processed through L identical attention layers to obtain an advanced input embedding h_i^L, facilitating a deeper understanding of relationships among the mission points. Each sublayer $l \in L$ includes a multi-head attention (MHA) layer, a skip connection layer, a feed-forward (FF) layer, and a batch normalization (BN) layer. For the MHA layer within each sublayer $l \in L$, three vectors, *Query*, *Key* and *Value* are calculated from the previous layer's node embedding h_i^{l-1}. The dimensions of *Query/Key*, *Value* are set as $d_q/d_k = d_v = \frac{d_h}{M}$, where $M = 8$ is the number of attention heads. For each attention head $j \in 1, 2, \ldots M$, attention scores Z_j^l are calculated using *Query*, *Key* and *Value*, which are concatenated together to produce the attention output $\text{MHA}(h_i^{l-1})$ of that layer. The calculations are shown here:

$$q_{i,j}^l = h_i^{l-1} W_{q,j}^l,\ k_{i,j}^l = h_i^{l-1} W_{k,j}^l,\ v_{i,j}^l = h_i^{l-1} W_{v,j}^l \quad (13)$$

$$Z_j^l = \text{softmax}\left(\frac{q_{i,j}^l {k_{i,j}^l}^T}{\sqrt{d_k}}\right) v_{i,j}^l \quad (14)$$

$$\text{MHA}(h_i^{l-1}) = \text{Concat}(Z_1^l, Z_2^l, \ldots, Z_j^l) \quad (15)$$

Here, $q_{i,j}^l, k_{i,j}^l$ and $v_{i,j}^l$ denote the *Query*, *Key* and *Value* for head j, respectively, and $W_{q,j}^l \in \mathbb{R}^{d_h \times d_q}, W_{k,j}^l \in \mathbb{R}^{d_h \times d_k}$ and $W_{v,j}^l \in \mathbb{R}^{d_h \times d_v}$ are the trainable parameter matrices in the MHA layer l. The attention output $\text{MHA}(h_i^{l-1})$ is then fed into a feed-forward layer (FF) with ReLU activation function to give the node embedding for that layer. Residual skip connections and Batch-Normalization (BN) are applied after both MHA and FF sublayers, as shown in the calculations below:

$$\hat{h}_i^l = \text{BN}(h_i^{l-1} + \text{MHA}(h_i^{l-1})) \quad (16)$$

$$h_i^l = \text{BN}(\hat{h}_i^l + FF(\text{ReLU}(\hat{h}_i^l))) \quad (17)$$

After $L = 3$ attention layers, the final node embedding h_i^L is obtained and subsequently forwarded to the decoder for further processing.

2) Decoder: In general, during each decision-making step t, the decoder determines the probability of selecting each available node as an action based on the encoder's node embedding h_i^L and a **context** vector, which provides insights into the current scenario state. The context vector h_t^c is constructed using three elements: the embedding of the agent's current position $h_{j,t}^L$, for $j \in p_t$, the agent's current fuel level f_t, and the final node embeddings h_i^L obtained from the encoder. To incorporate global information, we first compute the mean graph embedding as the average of final node embeddings:

$$\bar{h}_t = \frac{1}{n} \sum_{i=1}^{n} (h_i^L) \quad (18)$$

The context vector is then formed by concatenating the agent's current position embedding and fuel level, applying a linear projection, and combining this with a separate linear projection of the global graph embedding as shown:

$$h_t^c = \bar{h}_t W_g + \text{Cat}(h_{j,t}^L, f_t) W_c \quad (19)$$

Here, W_g, W_c are trainable parameters. To extract features from the current *age period* status of the mission points, we utilize an *age period* embedder. This embedder first performs layer normalization (LN) on the current *age period* status of the mission points and subsequently projects it linearly to obtain the *age period* embedding. Unlike previous works that use a fixed initial node embedding throughout the decoder's operations for computational efficiency, our approach combines the *age period* embedding with the initial node embedding to create a dynamic node embedding. This dynamic embedding is then utilized in subsequent steps to make decisions. The calculations are as follows:

$$\hat{\mathfrak{a}}_t = \text{LN}(\mathfrak{a}_t) \quad (20)$$

$$h_t^a = \hat{\mathfrak{a}}_t W_a + h_i^L \quad (21)$$

Here, W_a is a trainable parameter and $\mathfrak{a}_t$ represents current *age period*. The dynamic node embedding h_t^a is updated at each decision-making step as the *age period* status changes in the current state, capturing the essence of the mission points' visitation patterns. This helps to prioritize mission points based on the most recent information, leading to more informed and adaptive decision-making to effectively achieve the mission objective.

As the context vector and dynamic node embedding are constructed from the current state, the decoder employs a Multi-Head Attention (MHA) layer. In this layer, the context vector serves as the *Query*, and the dynamic node embedding acts as the *Key* and *Value* to compute the glimpse h_t^g, enabling the model to simultaneously aggregate multiple aspects of the current state. The calculations are as follows:

$$h_t^g = \text{MHA}(h_t^c,\ h_t^a W_k^g,\ h_t^a W_v^g) \quad (22)$$

Here, W_k^g, W_v^g are trainable parameter matrices. Once the glimpse h_t^g is obtained, it is used as the *Query*, and the dynamic node embedding h_t^a serves as the *Key* inside a single-head attention layer to compute their compatibility h_t. An important step here is to mask out infeasible actions based on the current scenario state. The infeasible actions include: 1) unreachable mission points at the current fuel level, 2) mission points that, if visited, would prevent the agent from reaching any refuel stop in the next step, 3) consecutive visits to the same mission point, and 4) performing recharging actions consecutively. These logical constraints are applied to determine infeasible actions, for which the compatibility is set to negative infinity. The compatibility between the *Query* q_t and *Key* k_t is calculated as follows:

$$q_t = h_t^g W_q, k_t = h_t^a W_k \quad (23)$$

$$h_t = \begin{cases} C_p \cdot \tanh(\frac{q_t {k_t}^T}{\sqrt{d_q}}) & \text{if feasible} \\ -\infty & \text{else.} \end{cases} \quad (24)$$

Here, W_q, W_k are trainable matrices, and $C_p = 10$ is a clipping parameter for better exploration. As the final step, the output probabilities for the actions are calculated using the softmax function on the compatibility:

$$\pi_\theta(a_t|s_t) = \text{softmax}(h_t) \tag{25}$$

Following the above process, the decoder sequentially determines actions to visit mission points or perform recharging until the mission is completed. We employ two decoding strategies: a greedy strategy that consistently selects the action with the highest probability at each decision-making step, and a sampling strategy that chooses actions based on their probabilities. During training, we adopt the sampling strategy to encourage better exploration. In the evaluation phase, we assess and compare the effectiveness of both strategies.

B. Training Method

The encoder-decoder transformer architecture is trained using a policy gradient approach to optimize cooperative UAV–UGV route generation. At a high level, the algorithm samples batches of routing instances, evaluates mission rewards, and updates policy network parameters to increase the likelihood of better trajectories in future iterations. A baseline network is maintained for comparison to reduce variance in policy updates and stabilize learning.

Algorithm 1 presents the detailed training procedure based on the REINFORCE policy gradient method [54]. The training involves two neural networks: the policy network π_θ calculates probability distributions over actions and selects actions by sampling, while the rollout baseline network π_ϕ has identical structure but selects actions greedily by choosing the highest probability action. At each training iteration, routes and corresponding rewards are calculated for a batch of instances, with baseline rewards obtained from greedy rollout in the baseline network (lines 4-13). The policy network parameters are updated according to the policy gradient algorithm (lines 14-16). After each epoch, the baseline network parameters ϕ are updated with those of the policy network if the baseline performs worse in a paired t-test (lines 17-19). This iterative approach ensures consistent policy network improvement through reinforcement learning while using the baseline network as a comparative measure to guide updates effectively.

Algorithm 1: Policy network training using REINFORCE algorithm.

Input: Policy network π_θ, Baseline network π_ϕ, epochs E, Number of batches N, batch size B, episode length T
Output: Trained policy network $\pi_{\theta'}$

1 **for** *epoch in* $1 \dots E$ **do**
2 Sample N batches from dataset
3 **for** *iteration in* $1 \dots N$ **do**
4 **for** *instance* b *in* $1 \dots B$ **do**
5 Initialize $s_{0,b}$ at $t = 0$
6 **while** $t < T$ **do**
7 Get action $a_{t,b} \sim \pi_\theta(a_{t,b}|s_{t,b})$
8 Obtain reward $r_{t,b}$ and $s_{t+1,b}$
9 $t = t + 1$
10 **end**
11 $\mathcal{R}_b = \frac{1}{T_m^2} \sum_{t=0}^{T_m} r_{t,b}$
12 Baseline reward $\mathcal{R}_b^\phi$ from greedy rollout with π_ϕ
13 **end**
14 Compute gradient:
$$\nabla_\theta J \leftarrow \frac{1}{B} \sum_{b=1}^{B} (\mathcal{R}_b - \mathcal{R}_b^\phi) \nabla_\theta \log \pi_\theta \left(s_{T,b} \mid s_{0,b}\right)$$
15 Update $\theta \leftarrow \theta - \alpha \nabla_\theta J$
16 **end**
17 **if** *OneSidedPairedTTest*$(\pi_\theta, \pi_\phi) < 0.05$ **then**
18 $\phi \leftarrow \theta$
19 **end**
20 **end**

VI. Results

To verify our proposed deep reinforcement learning framework's effectiveness, we conduct extensive computational experiments comparing against baseline methodologies and evaluating generalization robustness. We also examine online route planning applicability through dynamic planning and priority-driven surveillance case studies.

A. Dataset Details

We simulate the energy-constrained UAV–UGV cooperative persistent surveillance problem over a 20 km × 20 km area using a single UAV–UGV team. The UAV and UGV travel at constant speeds of $v_a = 10$ m/s and $v_g = 4.5$ m/s, respectively. The UAV has fuel capacity $F^a = 287.7$ kJ and follows the consumption model $\mathcal{P}^a = 0.0461(v_a)^3 - 0.5834(v_a)^2 - 1.8761 v_a + 229.6$, which provides a maximum flight time of 25 minutes at $v_a = 10$ m/s [55]. Both vehicles start from a common depot. The UAV visits off-road mission points $\mathcal{M}_a$ and road-network points $\mathcal{M}_g$, while the UGV is restricted to the road network and can recharge the UAV at any road-network location. The depot and UAV mission points are uniformly sampled within a 4 km radius of road-network points, while the road network G is fixed and the selected ground points are randomized. We evaluate three problem sizes: 15 UAV points with 5 ground points (**U15G5**), 30 UAV points with 10 ground points (**U30G10**), and 45 UAV points with 15 ground points (**U45G15**). The corresponding mission durations are 400, 600, and 700 minutes, respectively, allowing repeated visits to mission points in larger scenarios. For each size, 5,120,000 training instances are generated on-the-fly using 256 instances per batch, 200 batches, and 100 epochs. We train the model using the Adam optimizer with an initial learning rate of 10^{-4} and decay factor $\alpha = 0.995$, updated after each epoch as

$$lr_{\text{decayed}} = lr \times \alpha^{n_\text{epoch}}, \tag{26}$$

where lr is the current learning rate and n_epoch is the epoch index. Training on an RTX 2080 Ti GPU takes approximately 4, 6, and 9 minutes per epoch for U15G5, U30G10, and U45G15, respectively.

B. Comparison Evaluation

Given the complexity and specificity of the problem, standard benchmarks are not available, and deriving an exact solution is impractical due to the problem's intractability as the number of

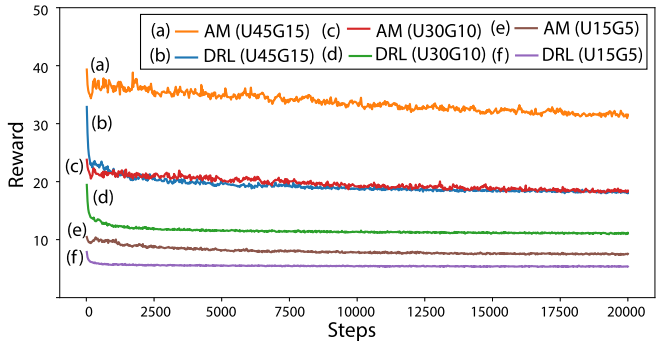


Fig. 5. Training reward curves across different problem sizes under DRL and AM policies.

mission points increases. Consequently, a multi-level or multi-echelon optimization strategy is a widely adopted approach for UAV-UGV cooperative routing. In persistent surveillance, as the mission points must be revisited continuously until the mission period concludes, the solution framework is often implemented in a receding horizon until a termination criterion is met. As outlined in the problem modeling (see Section IV), we employ a bi-level optimization strategy as the heuristic baseline method, utilizing three different metaheuristics: 1) **Guided Local Search (GLS)**, 2) **Tabu Search (TS)**, and 3) **Simulated Annealing (SA)** for comparison against the DRL policy. Additionally, we include a learning-based model, the **Attention Model (AM)** [36], as another baseline for further comparison. During the evaluation process, our deep reinforcement learning framework employs two types of decoding strategies: a) greedy decoding, where actions with the highest probability are chosen at each decision-making step, and b) sampling decoding, where $\mathbb{N}$ trajectories are sampled to form a solution pool, from which the best solution is selected. We set the number of samples ($\mathbb{N}$) to 1024 (DRL(1024)) and 10240 (DRL(10240)) in the evaluation instances. For a fair comparison between our DRL model and the AM model, we use the same set of hyperparameters and decoding strategies during evaluation. The evaluation process is conducted using an Nvidia Quadro P2200 GPU, and the persistent surveillance mission period is extended up to 1000 minutes for the testing instances. Given the longer mission period and the increased computational time required by the heuristic approach, we evaluate 30 test instances along with all baselines. All computations are implemented in Python.

Fig. 5 shows training curves for our DRL policy versus the baseline AM model across three problem sizes. Both models start with higher returns that converge to lower values during training. Our DRL policy demonstrates superior performance with lower objective function values compared to AM. As expected, objective values increase with problem size due to more mission points contributing to higher cumulative scores. The performance gap between DRL and AM models becomes more pronounced with larger problem sizes, indicating DRL's enhanced capability in managing larger instances.

Table I presents average objective values, runtimes, and optimality gaps across three problem sizes. Lower objective values achieved in shorter runtimes indicate superior solution quality.

TABLE I
COMPARISON EVALUATION OF THE DRL POLICY ACROSS PROBLEM SIZES

Method	U15G5			U30G10			U45G15		
	Obj	Gap (%)	Time (sec.)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)
AM(greedy)	7.72	191.3	2.2	25.92	296.9	5.6	42.69	266.8	6.1
DRL(greedy)	2.86	7.9	**1.6**	6.98	6.9	**2.9**	12.34	6.0	**3.3**
AM(1024)	3.77	42.3	6.7	11.51	76.3	9.8	21.99	88.9	9.5
DRL(1024)	2.67	0.8	7.3	6.60	1.1	10.5	11.77	1.1	12.8
AM(10240)	3.67	38.5	69.2	11.18	71.2	119.0	21.38	83.7	96.7
DRL(10240)	**2.65**	0.0	75.3	**6.53**	0.0	123.8	**11.64**	0.0	122.6
GLS	8.29	212.8	398.3	18.01	175.8	397.8	23.66	103.3	399.1
TS	3.10	17.0	366.0	10.50	60.8	394.2	20.37	75.0	399.2
SA	9.23	248.3	405.3	18.01	175.8	398.2	25.45	118.6	399.0

The optimality gap is calculated as the difference between the objective values and the best objective function value found, using the formula:

$$\text{optimality gap} = \frac{\text{Obj.} - \text{Obj.}_{\text{best}}}{\text{Obj.}_{\text{best}}} \times 100\% \qquad (27)$$

The results show that the proposed DRL policy consistently produces lower objective values across all problem sizes compared to the baselines. Among learning-based approaches, the DRL policy outperforms the AM policy across all problem sizes with both greedy and sampling decoding strategies. The sampling decoding strategy proves more effective than the greedy strategy, as it samples more solutions and selects the best from the solution pool. The optimality gap between the AM model and the DRL policy increases with problem size, ranging from 38.5% in smaller problems (U15G5) to 83.7% in larger problems (U45G15), highlighting the DRL policy's efficiency in managing larger scenarios. The DRL(10240) model achieves the best performance, producing the minimum objective scores across all problem sizes compared to the baselines. The AM model shows significant variance between its performance with greedy and sampling decoding strategies, with AM(greedy) performing the worst across all methods due to poor convergence during training. In contrast, the DRL policy shows less variation between greedy and sampling decoding, indicating good convergence of the network. Among the heuristic-based models, although the Guided Local Search (GLS) and Simulated Annealing (SA) heuristics perform worse than the learned models, the Tabu Search (TS) heuristic outperforms the baseline AM model and provides comparable performance to that of DRL policy. The optimality gap between DRL(10240) and TS heuristics ranges from 17% in smaller problem sizes to 75% in larger problem sizes.

For computational efficiency, DRL with greedy decoding yields the shortest runtimes. Sampling decoding requires longer computation due to solution sampling overhead. DRL generally requires slightly more runtime than AM except with greedy decoding, and runtime increases linearly with problem size for both policies. The heuristic-based methods have runtimes ranging from 366 to 405 seconds, with TS heuristics being the fastest among the heuristics but still significantly slower than DRL(10240). Specifically, TS heuristics takes 5 times longer than DRL(10240) for small-scale problems, 3.5 times longer for medium-scale problems, and 3.2 times longer for large-scale problems. Considering both solution quality and

TABLE II
COMPARISON EVALUATION OF THE DRL POLICY ACROSS LARGER PROBLEM SIZES

Method	U60G20			U75G25		
	Obj	Gap (%)	Time (sec.)	Obj	Gap (%)	Time (sec)
AM(greedy)	60.86	352.2	10.04	79.93	324.3	11.26
DRL(greedy)	15.19	12.9	**7.09**	19.88	5.5	**7.38**
AM(1024)	28.88	114.6	19.9	40.38	114.3	18.62
DRL(1024)	13.75	2.2	28.75	19.04	1.1	28.1
AM(10240)	28.16	109.2	192.1	39.48	109.6	179.46
DRL(10240)	**13.46**	0.0	288.1	**18.84**	0.0	285.2
GLS	42.35	214.6	801.9	49.25	161.4	807.2
TS	32.32	140.1	801.7	41.41	119.8	807.9
SA	42.90	218.7	802.1	49.78	164.2	807.1

runtime, DRL(greedy) achieves the optimal balance between these aspects. Route animations for different methods across problem scenarios are available at our website[1].

C. Generalization

To analyze the generalization capability of the proposed DRL framework, we examine different testing instances by modifying two aspects of the problem scenarios: 1) increasing the problem size beyond the original training size and 2) creating problem scenarios with three different distributions of mission points to mimic diverse operational environments.

1) Larger Problem Size: In the first experiment, we assess the model's ability to handle larger problem scenarios than its original training size by creating 30 test instances of two larger sizes. Firstly, we increase the number of mission points to a) 60 UAV points and 20 ground points (U60G20) and b) 75 UAV points and 25 ground points (U75G25). Secondly, we extend the UAV points' spread radius to 6 km from the road network. Thirdly, the road network has also been expanded to have more ground points. We also extend the persistent surveillance mission period to 2000 minutes for these scenarios. The model's performance is compared against the baselines on these U60G20 and U75G25 test instances, as listed in Table II. The DRL(10240) policy achieves the lowest objective score among the methods. Our DRL policy demonstrates better generalization than the AM model in terms of solution quality with both greedy and sampling decoding strategies, showing an optimality gap of approximately 110%. Overall, learning-based methods (DRL and AM) produce lower scores than heuristic methods, except for AM(greedy), which performs the worst in solution quality due to poor convergence. Among heuristics, the Tabu Search (TS) performs best with an optimality gap of approximately 120-142%. With growing problem sizes, the total objective score increases due to the cumulative contribution from the more number of mission points. The solution runtime also increases with larger problem sizes; however, the learned models produce faster solutions than heuristic methods. Our DRL policy with the sampling decoding strategy takes longer runtime than the AM model due to additional computation; however, DRL(greedy) emerges as the fastest method, with approximately 400 times faster runtime than heuristic solutions.

[1]https://sites.google.com/view/ecucps

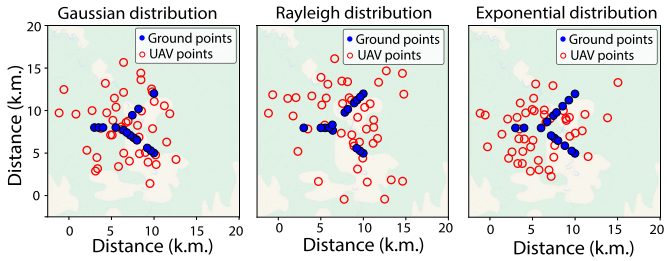


Fig. 6. Instances of scenarios with different UAV points distributions around the road network.

2) Mission Points Distribution Variations: In our second generalization experiment, we test 30 scenarios with three different distributions of UAV mission points around the road network: a) Gaussian distribution, b) Rayleigh distribution, and c) Exponential distribution. In the Gaussian distribution, UAV points cluster around a central point, with the frequency decreasing symmetrically as the distance from the center increases. This distribution is ideal for surveillance needs concentrated around a specific location or feature, such as a building or central urban area, where events are most likely to occur. In the Rayleigh distribution, the density of UAV mission points is low near the central point and increases up to a certain radial distance, beyond which it decreases again. These scenarios are useful for situations where the area immediately around a central point does not require surveillance, but the surrounding area does, reflecting a buffer zone around a secure site. In the Exponential distribution, UAV mission points are spread such that the probability of occurrence decreases exponentially from a central point. This results in a high density of points close to the origin, with rapidly fewer points as the distance increases. This distribution is suitable for scenarios where the importance or risk decreases sharply with distance from a point of interest. Fig. 6 shows example scenarios from each type of distribution. These varied distributions help evaluate the model's ability to generalize across different real-world scenarios, ensuring adaptability to different mission requirements. We test our proposed model along with baselines on three problem sizes (U15G5, U30G10, U45G15) of each distribution for a persistent surveillance period of 1000 minutes, and the results are represented in Table III.

Across the three distributions, our DRL(10240) policy consistently outperforms the baseline models. However, the Tabu Search (TS) heuristic model offers comparable performance to our DRL(greedy) model, with an optimality gap of 9.5-69.5% relative to DRL(10240) across small (U15G5) to large-scale (U45G15) scenarios. The TS heuristic model surpasses the AM models in all problem scenarios. Conversely, the GLS and SA heuristics demonstrate poorer performance among the heuristic approaches and AM(greedy) performing the worst in terms of solution quality. Generally, the objective scores are lowest in the exponential distribution due to the centralized clustering of mission points. They increase from the Gaussian distribution to the highest in the Rayleigh distribution, attributed to the wider radial spread of mission points. The greedy decoding strategy achieves faster runtimes than the sampling decoding strategy

TABLE III
COMPARISON EVALUATION OF THE DRL POLICY ACROSS DIFFERENT MISSION POINT DISTRIBUTIONS

Method	Gaussian distribution									Rayleigh distribution									Exponential distribution								
	U15G5			U30G10			U45G15			U15G5			U30G10			U45G15			U15G5			U30G10			U45G15		
	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)	Obj	Gap (%)	Time (sec)
AM(greedy)	8.35	186.0	2.3	25.89	249.4	4.4	42.43	215.5	6.5	10.88	222.8	2.0	27.17	226.2	5.5	41.62	172.7	5.7	8.85	205.2	3.1	25.79	248.0	3.9	44.28	249.8	6.5
DRL(greedy)	3.23	10.6	**1.8**	8.11	9.4	**2.7**	14.72	9.4	**3.4**	3.68	9.2	**1.5**	9.16	10.0	**2.5**	17.36	13.8	**3.0**	3.12	7.6	**2.2**	8.07	8.9	**2.3**	13.67	8.0	**3.5**
AM(1024)	4.24	45.2	6.5	12.56	69.5	9.3	23.71	76.3	10.2	5.12	51.9	6.4	14.16	70.0	9.8	25.79	69.0	10.7	4.20	44.8	11.0	12.70	71.4	9.9	22.78	79.9	11.7
DRL(1024)	2.96	1.4	6.4	7.52	1.5	10.7	13.70	1.9	12.3	3.40	0.9	6.0	8.45	1.4	9.3	15.53	1.8	12.3	2.92	0.7	11.8	7.50	1.2	10.9	12.79	1.0	13.7
AM(10240)	4.11	40.8	67.0	12.19	64.5	94.2	23.17	72.3	102.3	4.94	46.6	68.5	13.65	63.9	97.1	25.26	65.5	87.0	4.06	40.0	103.2	12.21	64.8	99.1	22.22	75.5	113.4
DRL(10240)	**2.92**	0.0	82.3	**7.41**	0.0	135.7	**13.45**	0.0	135.2	**3.37**	0.0	63.5	**8.33**	0.0	94.2	**15.26**	0.0	115.7	**2.90**	0.0	74.8	**7.41**	0.0	95.8	**12.66**	0.0	132.7
Heu GLS	8.47	190.1	394.4	17.90	141.6	398.1	27.77	106.5	399.9	9.20	173.0	397.9	18.96	127.6	397.8	28.04	83.7	399.7	9.00	210.3	315.9	17.76	139.7	397.6	25.68	102.8	399.3
Heu TS	3.25	11.3	369.2	11.84	59.8	395.5	22.15	64.7	400.2	3.69	9.5	372.2	11.30	35.7	393.4	22.05	44.5	399.8	3.30	13.8	370.2	11.95	61.3	395.3	21.46	69.5	399.3
Heu SA	10.24	250.7	399.9	20.20	172.6	399.7	29.11	116.4	399.9	10.81	220.8	405.5	20.42	145.1	399.8	30.48	99.7	400.0	8.08	178.6	397.9	18.90	155.1	398.1	28.23	123.0	399.6

due to reduced computation. DRL(greedy) strikes an optimal balance between time and performance, producing solutions of similar quality to TS heuristics but in significantly less time, 200 times faster in U15G5, 100 times faster in U30G10, and 65 times faster in U45G15 scenarios.

D. Case Study Analysis: Hurricane Harvey 2017

To evaluate the real-world applicability of our proposed framework and gain a deeper understanding of the interaction between the UAV and UGV in routing, recharging, and their impact on persistent surveillance mission performance, we apply our trained model to a case study based on the Hurricane Harvey disaster of 2017 [56]. Hurricane Harvey, which struck in August 2017, was one of the most devastating storms in U.S. history, causing catastrophic flooding in Texas and resulting in over $125 billion in damages. The hurricane's unprecedented rainfall, exceeding 60 inches in some areas, displaced thousands and led to lasting environmental and economic challenges, making it an ideal testbed for our case study. To construct a realistic scenario based on this disaster, we utilize the ArcGIS disaster map layer [57] of Hurricane Harvey alongside the USA population density map [58]. The map is divided into square grids, each covering a sensor footprint of 1600 square meters, with a focus on highly populated regions around Houston. We identify 20 UAV-only mission points and 10 ground points from the road network as surveillance targets, over a 1000-minute surveillance period, particularly within flood hazard and high water extent risk zones (see Fig. 7(ia)). Given the superior performance of the sampling decoding strategy for quality solutions, we select DRL(10240) as the DRL model and Tabu Search (TS) heuristics as the heuristic-based approach for the case study evaluation.

After conducting surveillance, we analyze the impact on individual mission points and their spatial significance by plotting the mission point-wise contributed score metric using a color map. Darker colors indicate greater contribution, reflecting longer intervals between visits or higher *age periods* observed at those mission points. As shown in Fig. 7(i), mission points located farther from the road network and isolated from neighboring points exhibit higher *score metrics*. This trend is particularly evident in the performance of the TS heuristics. The heuristic method is applied using a receding horizon approach, where each planning horizon optimizes the route sortie by solving the O-EVRPTW based on the current *age period* status of mission points. However, the optimal solution at a given planning instance may not remain optimal over the entire mission period, leading to the *horizon effect*. This effect occurs because the planning horizon only considers a limited timeframe, which can result in suboptimal decisions for mission points that fall outside this immediate planning window. The UAV tends to visit more closely clustered mission points within a region to achieve a higher cumulative reward from the O-EVRPTW during its planning horizon. Consequently, the far isolated points experience less frequent visits, resulting in higher *age periods* and contributing significantly to the score metric. This suboptimal visitation pattern is a direct consequence of the *horizon effect*, as the receding horizon approach prioritizes immediate rewards over long-term efficiency, thereby neglecting isolated mission points. The *horizon effect* is mitigated in DRL planning, as the entire cooperative route is evaluated at the end of the mission, and its performance is enhanced through a reward-based feedback mechanism. Thus, extreme mission points experience shorter *age periods* under the DRL policy compared to the TS heuristics. We also calculate the maximum *age period* observed at each mission point during the mission, as depicted in Fig. 7(ii). The TS heuristics method results in longer maximum *age periods* for mission points located far from road network points. Throughout the mission, we monitor the maximum and average *age periods* across all mission points, and objective score metric as shown in Fig. 7(iii), where the DRL policy consistently outperforms the TS heuristics. At any timestep t, the maximum and average *age periods* can be defined as follows:

$$\text{Max. Age} = \max_{j=1,..,n} (\mathfrak{a}_t^j) \tag{28}$$

$$\text{Avg. Age} = \frac{1}{n} \sum_{i=1}^{n} (\mathfrak{a}_t^j) \tag{29}$$

here, $\mathfrak{a}_t^j$ is the *age period* of mission point m_j at timestep t.

E. Dynamic Planning

Our case study also investigates the potential for dynamic planning in persistent surveillance, particularly when new mission points emerge randomly during the routing process. We

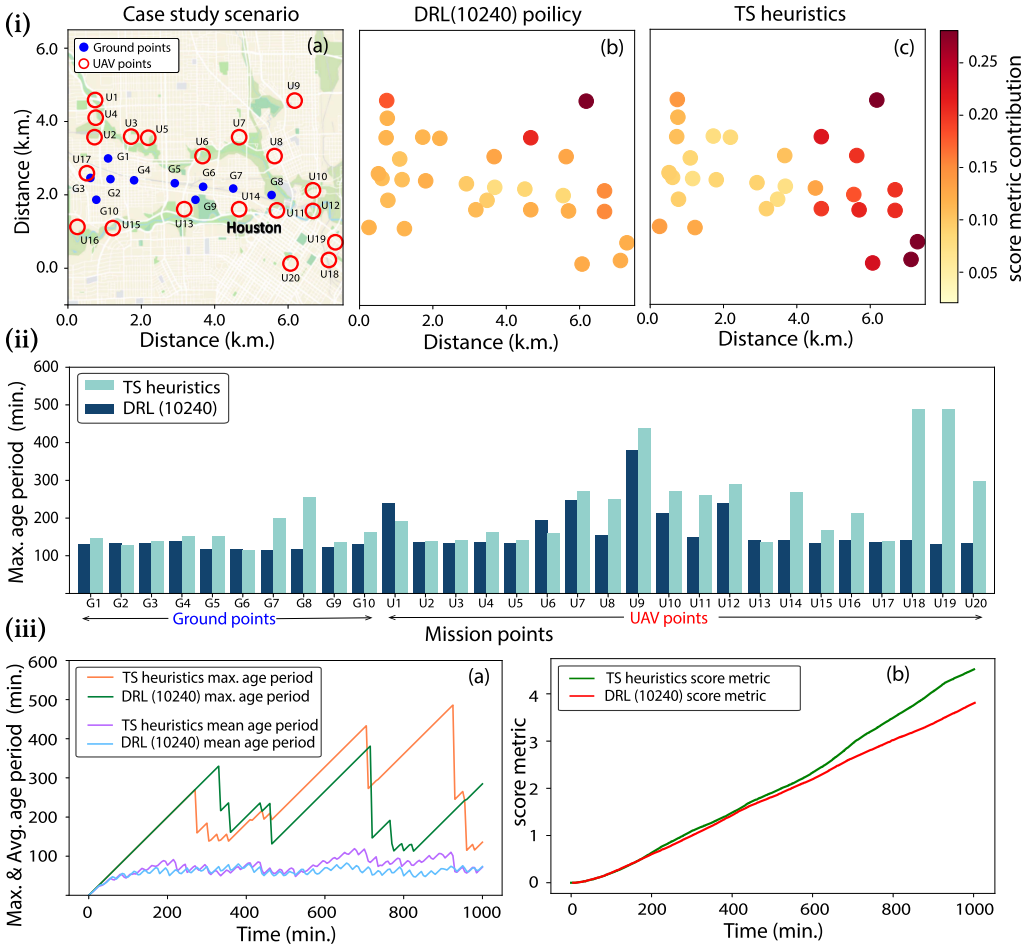


Fig. 7. (Best viewed in color) Analysis of objective score metric and *age period* metrics across mission points in the Hurricane Harvey 2017 case study. (i) Contribution to the objective score metric from individual mission points: (a) Case study scenario illustrating ground (blue) and UAV (red) mission points. (b) Heatmap showing the individual score contributions of mission points from the DRL(10240) policy. (c) Heatmap showing the individual score contributions of mission points from the TS heuristics policy. (ii) Maximum *age period* observed at each mission point, comparing DRL(10240) and TS heuristics policies. (iii) Comparison of *age period* metrics and objective score metric over time: (a) Maximum and mean *age period* across mission points during mission progression for both the DRL(10240) and TS heuristics policies. (b) Objective score metric progression throughout the mission, as obtained from both the DRL(10240) and TS heuristics policies.

utilize our trained DRL policy to manage these dynamic changes for online planning. These new mission points are assumed to be located outside the road network, making them accessible only to the UAV. A critical assumption is that the UAV and UGV can exchange information exclusively during their rendezvous. Following the initial planning to establish the primary route, the trained policy updates its encoder space at each rendezvous to incorporate any new mission points, adjusting its actions as necessary. We choose the DRL(greedy) method due to its rapid execution time, which makes it highly suitable for online planning. We conduct a persistent surveillance mission lasting 1000 minutes, with the assumption that new mission points will appear at random locations within the first 200 minutes. This approach ensures that the UAV has sufficient time to visit the mission points multiple times during the remainder of the mission, enabling a comprehensive evaluation of our policy's effectiveness.

To assess the robustness of our model in response to the appearance of new mission points, we calculate the score metric and the maximum *age period* for both the initial route and the dynamically adjusted route. We have conducted 30 trials of dynamic planning with an average of 5 newly appeared mission points, and the results detailed in Table IV show little deviations in objective score metric and maximum *age period* across all mission points after addition of new mission points. These findings underscore the efficacy of our DRL policy in dynamically adapting to new mission points in persistent surveillance scenarios. Fig. 8 illustrates the temporal adaptation of the UAV-UGV route obtained from the DRL policy as they adjust their initial plan to accommodate the newly emerged mission points.

TABLE IV
EFFECT OF THE DYNAMIC PLANNING FOR NEWLY ADDED RANDOM MISSION POINTS IN THE CASE STUDY SCENARIO

Routes	score metric	Max. Age (minutes)
Initial route	3.92	378
Modified routes	5.51	523

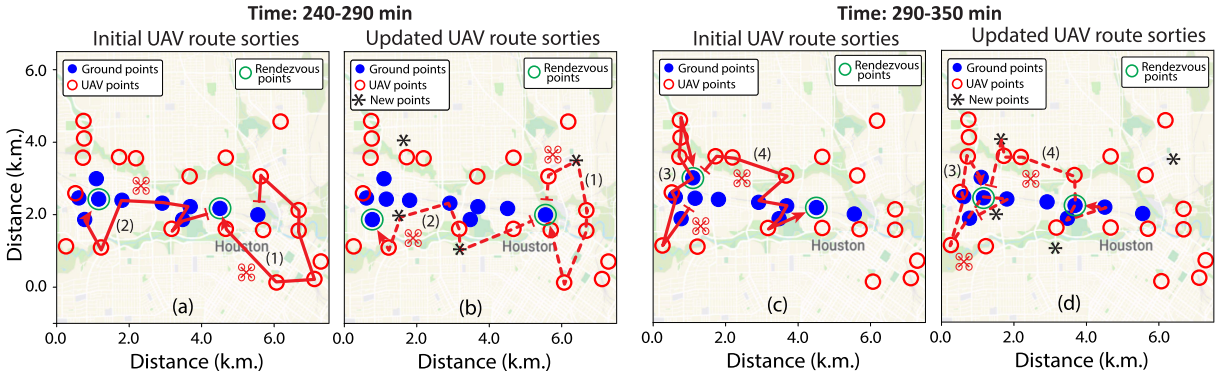


Fig. 8. Dynamic route planning for newly appearing mission points (a) Initial UAV route sorties during 240–290 minutes in the case study scenario (b) Updated UAV route sorties during 240–290 minutes in response to newly added mission points (c) Initial UAV route sorties during 290–350 minutes (d) Updated UAV route sorties during 290–350 minutes in response to newly added mission points. The route animation can be found at our website.

F. Priority Driven Persistent Surveillance

We extend our case study to illustrate the applicability of the DRL policy in a priority-driven persistent surveillance scenario. In this context, target mission points are assigned different weights based on their required visit frequency, which allows for prioritizing them accordingly. To demonstrate this, we modify our case study to assign higher weightage to five mission point locations that observed fewer visits (indicated by a higher maximum *age period* in Fig. 7(ii), represented by a deeper red color in Fig. 7(ib) and lower weightage to another five mission points that had higher visitation rates (indicated by a lower maximum *age period*, represented by a lighter yellow color in Fig. 7(ii) and Fig. 7(ib) respectively). The objective score metric function is also adjusted as follows:

$$\mathbb{S} = \frac{1}{\alpha}\sum_{k=1}^{n}\sum_{q=1}^{p} w^k \left(t_q^k - t_{q-1}^k\right)^2 = \frac{1}{\alpha}\sum_{k=1}^{n}\sum_{q=1}^{p} w^k (\mathfrak{a}_q^k)^2 \quad (30)$$

Here, w^k is the weight for mission point m_k. For high-priority mission points, we set w^k= 1.5, for low-priority mission points, $w^k = 0.5$, and for normal mission points, $w^k = 1$. The DRL policy decides to visit mission points based on the **context** vector that accounts for the *age period* status of the mission points at each decision-making step. To adapt our DRL policy, originally trained with uniform weightage ($w^k = 1$) for all mission points, to a priority-based scenario, we adjust the *age period* status of the mission points between two timesteps by multiplying the elapsed time with an increment factor $\mathcal{F}$ and adding it to the previous *age period* status, as follows:

$$\mathfrak{a}_{t+1}^k = \mathfrak{a}_t^k + t_{\text{elapsed}} * \mathcal{F} \quad (31)$$

$$\mathcal{F} = 1 + (w^k - 1)\mathcal{S} \quad (32)$$

The increment factor $\mathcal{F}$ is a function of a hyperparameter $\mathcal{S}$, which adjusts the rate of *age period* increment based on the weights of the mission points. For high-priority mission points ($w^k > 1$), $\mathcal{F}$ is greater than 1, leading to a more rapid increase in the *age period* compared to non-weighted points. Conversely, for lower-weighted mission points ($w^k < 1$), the *age period* increment will be less. This modified *age period* status allows the context vector to effectively capture the prioritization of mission points, ensuring they are visited accordingly. To optimize the hyperparameter $\mathcal{S}$, we conduct trials across a range of $\mathcal{S}$ values and select the optimal value that minimizes the objective score metric, as illustrated in Fig. 9.

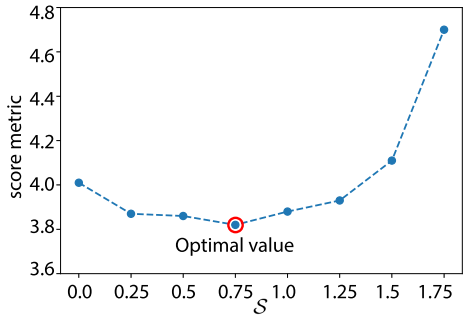


Fig. 9. Variation in the objective score metric for the prioritized persistent surveillance scenario based on the hyperparameter $\mathcal{S}$.

We have executed the routing process on the given scenario using the defined priority weights and the increment factor $\mathcal{F}$ with the optimal $\mathcal{S} = 0.75$. The maximum *age period* metrics across the mission points, as derived from the routing process, is illustrated in Fig. 10. Compared to the uniformly weighted case, the results show a significant reduction in the maximum *age periods* for high-priority mission points, while low-priority nodes experience a substantial increase in their maximum *age periods*. This demonstrates the policy's effectiveness in prioritizing mission points based on their assigned weights.

Both dynamic planning and priority-driven surveillance can be integrated for newly appearing mission points requiring immediate attention. Dynamic event detection triggers replanning while prioritized weightage ensures urgent points receive more frequent UAV visits. This approach proves particularly valuable in disaster management where critical sites demand more monitoring than less critical locations.

G. Hardware Deployment Considerations

Our DRL framework generates cooperative UAV–UGV routes for persistent surveillance missions. While simulation results show effectiveness, real-world deployment faces additional

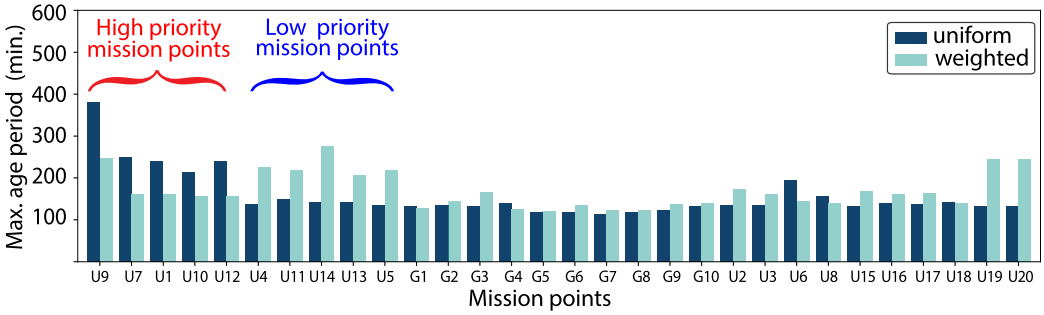


Fig. 10. Maximum *age periods* at individual mission points in the case study scenario, comparing uniform surveillance to priority-weighted surveillance. In priority-driven surveillance, high-priority mission points observe a decrease in maximum *age periods*, while low-priority points see an increase in *age period* values.

challenges. Communication disruptions in disaster zones require robust coordination strategies, such as aerial P2P backbone networks [59], to maintain UAV–UGV connectivity. Charging operations must account for battery degradation and delays, with UGV charging station strategies [60] providing guidance for realistic modeling. Though our framework minimizes *age periods* under energy constraints, adaptive scheduling techniques [61] offer complementary approaches for enhanced efficiency. The computational demands of DRL-based planning may require edge computing integration for real-time decision-making on resource-constrained platforms. These considerations highlight key directions for extending our framework to robust real-world UAV–UGV deployment.

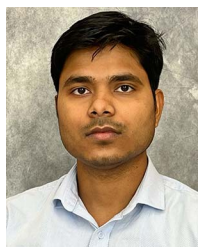
VII. Conclusion and Future Work

This study addresses energy-aware UAV-UGV cooperative persistent surveillance for continuous monitoring in disaster-stricken areas. The problem involves an energy-constrained UAV and UGV visiting predefined mission points, with the UAV periodically recharged by the UGV acting as a mobile depot. We propose a deep reinforcement learning (DRL) framework using a transformer network with attention layers to optimize mission point visitation and UAV-UGV recharging rendezvous. The encoder generates node embeddings from mission coordinates, while the decoder determines actions using these embeddings and contextual state. To minimize the visit-interval period *age period* of mission points, the decoder dynamically updates the *age period* embedding at each decision point, prioritizing mission points with higher age periods. Evaluation against conventional multi-stage optimization strategies and existing learning-based models demonstrates: 1) Our DRL framework outperforms baselines across all problem sizes with lower objective scores and faster runtimes. 2) Superior generalization capability in larger scenarios and varying mission point distributions. 3) Successful evaluation on the Texas Hurricane Harvey 2017 scenario in simulation, demonstrating adaptability to dynamic changes. 4) Effective implementation in priority-driven surveillance scenarios without retraining. Future work will focus on extending the framework to multiple UAVs and UGVs, incorporating stochastic variations in vehicle dynamics, and performing experimental validation in both physics-based simulations and hardware, thereby advancing the practical deployment of persistent surveillance in dynamic environments.

References

[1] J. Rajan, S. Shriwastav, A. Kashyap, A. Ratnoo, and D. Ghose, "Disaster management using unmanned aerial vehicles," in *Unmanned Aerial Systems*. Amsterdam, The Netherlands: Elsevier, 2021, pp. 129–155.
[2] M. Erdelj, E. Natalizio, K. R. Chowdhury, and I. F. Akyildiz, "Help from the sky: Leveraging UAVs for disaster management," *IEEE Pervasive Comput.*, vol. 16, no. 1, pp. 24–32, Jan.–Mar. 2017.
[3] S. K. Chaturvedi, R. Sekhar, S. Banerjee, and H. Kamal, "Comparative review study of military and civilian unmanned aerial vehicles (UAVs)," *INCAS Bull.*, vol. 11, no. 3, pp. 181–182, 2019.
[4] M. Raffetto, "Unmanned aerial vehicle contributions to intelligence, surveillance, and reconnaissance missions for expeditionary operations," Ph.D. dissertation, Monterey, California. Naval Postgraduate School, Monterey, CA, USA, 2004.
[5] F. Yamazaki, T. Matsuda, S. Denda, and W. Liu, "Construction of 3D models of buildings damaged by earthquakes using UAV aerial images," in *Proc. 10th Pacific Conf. Earthq. Eng. Building Earthq. Resilient Pacific*, 2015, vol. 204.
[6] G. Salmoral, M. Rivas Casado, M. Muthusamy, D. Butler, P. P. Menon, and P. Leinster, "Guidelines for the use of unmanned aerial systems in flood emergency response," *Water*, vol. 12, no. 2, 2020, Art. no. 521.
[7] K. Gupta, S. Bansal, and R. Goel, "Uses of drones in fighting COVID-19 pandemic," in *Proc. 10th Int. Conf. Syst. Model. Advance. Res. Trends*, 2021, pp. 651–655.
[8] X. Lin, Y. Yazıcıoğlu, and D. Aksaray, "Robust planning for persistent surveillance with energy-constrained UAVs and mobile charging stations," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 4157–4164, Apr. 2022.
[9] K. Yu, J. M. O'Kane, and P. Tokekar, "Coverage of an environment using energy-constrained unmanned aerial vehicles," in *Proc. Int. Conf. Robot. Automat.*, 2019, pp. 3259–3265.
[10] P. Maini and P. Sujit, "On cooperation between a fuel constrained UAV and a refueling UGV for large scale mapping applications," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2015, pp. 1370–1377.
[11] K. Sundar, S. Venkatachalam, and S. Rathinam, "Formulations and algorithms for the multiple depot, fuel-constrained, multiple vehicle routing problem," in *Proc. Amer. Control Conf.*, 2016, pp. 6489–6494.
[12] C. C. Murray and A. G. Chu, "The flying sidekick traveling salesman problem: Optimization of drone-assisted parcel delivery," *Transp. Res. Part C, Emerg. Technol.*, vol. 54, pp. 86–109, 2015.
[13] Z. Wang and J.-B. Sheu, "Vehicle routing problem with drones," *Transp. Res. Part B, Methodological*, vol. 122, pp. 350–364, 2019.
[14] Z. Tang, W.-J. V. Hoeve, and P. Shaw, "A study on the traveling salesman problem with a drone," in *Proc. Integration Constraint Program., Artif. Intell., Operations Res., 16th Int. Conf.*, Thessaloniki, Greece: Springer, Jun. 2019, pp. 557–564.
[15] J. Scherer and B. Rinner, "Persistent multi-UAV surveillance with energy and communication constraints," in *Proc. IEEE Int. Conf. Automat. Sci. Eng.*, 2016, pp. 1225–1230.
[16] E. Angün and U. Dündar, "Intelligent systems for disaster management: Unmanned aerial vehicles' surveillance problem with energy consumption constraints," in *Proc. Intell. Fuzzy Techn. Big Data Analytics Decis. Making, Proc. Conf.*, Istanbul, Turkey, 2020, pp. 1269–1278.

[17] T. Calamoneri, F. Corò, and S. Mancini, "Management of a post-disaster emergency scenario through unmanned aerial vehicles: Multi-depot multi-trip vehicle routing with total completion time minimization," *Expert Syst. Appl.*, vol. 251, 2024, Art. no. 123766.
[18] N. Nigam, "The multiple unmanned air vehicle persistent surveillance problem: A review," *Machines*, vol. 2, no. 1, pp. 13–72, 2014.
[19] N. Michael, E. Stump, and K. Mohta, "Persistent surveillance with a team of MAVs," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2011, pp. 2708–2714.
[20] S. K. K. Hari, S. Rathinam, S. Darbha, K. Kalyanam, S. G. Manyam, and D. Casbeer, "Optimal UAV route planning for persistent monitoring missions," *IEEE Trans. Robot.*, vol. 37, no. 2, pp. 550–566, Apr. 2021.
[21] D. Cattaruzza, N. Absi, and D. Feillet, "Vehicle routing problems with multiple trips," *4OR*, vol. 14, pp. 223–259, 2016.
[22] F. Ropero, P. Muñoz, and M. D. R-Moreno, "TERRA: A path planning algorithm for cooperative UGV–UAV exploration," *Eng. Appl. Artif. Intell.*, vol. 78, pp. 260–272, 2019.
[23] M. Chen, Y. Chen, Z. Chen, and Y. Yang, "Path planning of UAV-UGV heterogeneous robot system in road network," in *Proc. Intell. Robot. Appl.*, Shenyang, China, Aug. 2019, pp. 497–507.
[24] M. Zhang, H. Liang, and P. Zhou, "Cooperative route planning for fuel-constrained UGV-UAV exploration," in *Proc. IEEE Int. Conf. Unmanned Syst.*, 2022, pp. 1047–1052.
[25] S. Seyedi, Y. Yazicioğlu, and D. Aksaray, "Persistent surveillance with energy-constrained UAVs and mobile charging stations," *IFAC-PapersOnLine*, vol. 52, no. 20, pp. 193–198, 2019.
[26] P. Maini, K. Sundar, M. Singh, S. Rathinam, and P. Sujit, "Cooperative aerial–ground vehicle route planning with fuel constraints for coverage applications," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 6, pp. 3016–3028, Dec. 2019.
[27] N. Nigam and I. Kroo, "Persistent surveillance using multiple unmanned air vehicles," in *Proc. IEEE Aerosp. Conf.*, 2008, pp. 1–14.
[28] K. Chour et al., "A reactive energy-aware rendezvous planning approach for multi-vehicle teams," in *Proc. IEEE 18th Int. Conf. Automat. Sci. Eng.*, 2022, pp. 537–542.
[29] A. B. Asghar et al., "Risk-aware recharging rendezvous for a collaborative team of UAVs and UGVs," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2023, pp. 5544–5550.
[30] Y. Kaempfer and L. Wolf, "Learning the multiple traveling salesmen problem with permutation invariant pooling networks," 2018, *arXiv:1803.09621*.
[31] Z. Wang and M. Gombolay, "Learning scheduling policies for multi-robot coordination with graph attention networks," *IEEE Robot. Automat. Lett.*, vol. 5, no. 3, pp. 4509–4516, Jul. 2020.
[32] Q. Sykora, M. Ren, and R. Urtasun, "Multi-agent routing value iteration network," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9300–9310.
[33] S. Paul, P. Ghassemi, and S. Chowdhury, "Learning scalable policies over graphs for multi-robot task allocation using capsule attention networks," in *Proc. Int. Conf. Robot. Automat.*, 2022, pp. 8815–8822.
[34] O. Vinyals, M. Fortunato, and N. Jaitly, "Pointer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, vol. 28, pp. 2692–2700.
[35] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in *Proc Int. Conf. Learn. Representation*, 2015.
[36] W. Kool, H. Van Hoof, and M. Welling, "Attention, learn to solve routing problems," in *Proc. Int. Conf. Learn. Representations*, 2019.
[37] J. Li et al., "Deep reinforcement learning for solving the heterogeneous capacitated vehicle routing problem," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13572–13585, Dec. 2022.
[38] G. Wu, M. Fan, J. Shi, and Y. Feng, "Reinforcement learning based truck-and-drone coordinated delivery," *IEEE Trans. Artif. Intell.*, vol. 4, no. 4, pp. 754–763, Aug. 2023.
[39] Q. Ma, S. Ge, D. He, D. Thaker, and I. Drori, "Combinatorial optimization by graph pointer networks and hierarchical reinforcement learning," 2019, *arXiv:1911.04936*.
[40] Q. Cappart, D. Chételat, E. B. Khalil, A. Lodi, C. Morris, and P. Veličković, "Combinatorial optimization and reasoning with graph neural networks," *J. Mach. Learn. Res.*, vol. 24, no. 130, pp. 1–61, 2023.
[41] C. K. Joshi, T. Laurent, and X. Bresson, "An efficient graph convolutional network technique for the travelling salesman problem," 2019, *arXiv:1906.01227*.
[42] M. Fan et al., "Deep reinforcement learning for UAV routing in the presence of multiple charging stations," *IEEE Trans. Veh. Technol.*, vol. 72, no. 5, pp. 5732–5746, May 2023.
[43] H. Bana, M. Mishra, S. Sarkar, S. Sanjeevi, S. PB, and K. Sundar, "Deep reinforcement learning-based approach for a single vehicle persistent surveillance problem with fuel constraints," 2024, *arXiv:2404.06423*.
[44] S. Ramasamy et al., "Heterogenous vehicle routing: Comparing parameter tuning using genetic algorithm and Bayesian optimization," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2022, pp. 104–113.
[45] M. S. Mondal, S. Ramasamy, and P. Bhounsule, "An attention-aware deep reinforcement learning framework for UAV-UGV collaborative route planning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2024, pp. 13687–13694.
[46] M. S. Mondal et al., "Cooperative multi-agent planning framework for fuel constrained UAV-UGV routing problem," *J. Intell. Robot. Syst.*, vol. 111, 2025, Art. no. 12.
[47] M. S. Mondal et al., "Optimizing fuel-constrained UAV-UGV routes for large scale coverage: Bilevel planning in heterogeneous multi-agent systems," in *Proc. Int. Symp. Multi-Robot Multi-Agent Syst.*, 2023, pp. 114–120.
[48] Google, "Google OR-tools," 2021. Accessed: Feb. 2, 2021. [Online]. Available: https://developers.google.com/optimization
[49] S. Ramasamy, J.-P. F. Reddinger, J. M. Dotterweich, M. A. Childers, and P. A. Bhounsule, "Coordinated route planning of multiple fuel-constrained unmanned aerial systems with recharging on an unmanned ground vehicle for mission coverage," *J. Intell. Robotic Syst.*, vol. 106, no. 1, 2022, Art. no. 30.
[50] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, vol. 30, pp. 6000–6010.
[51] L. Xin, W. Song, Z. Cao, and J. Zhang, "Step-wise deep learning models for solving routing problems," *IEEE Trans. Ind. Inform.*, vol. 17, no. 7, pp. 4861–4871, Jul. 2021.
[52] J. Yu, J. Li, Z. Yu, and Q. Huang, "Multimodal transformer with multi-view visual representation for image captioning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4467–4480, Dec. 2020.
[53] F. Sun et al., "BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer," in *Proc. 28th ACM Int. Conf. Inf. Knowl. Manage.*, 2019, pp. 1441–1450.
[54] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," *Mach. Learn.*, vol. 8, pp. 229–256, 1992.
[55] A. M. Hurwitz, J. M. Dotterweich, and T. A. Rocks, "Mobile robot battery life estimation: Battery energy use of an unmanned ground vehicle," *Proc. SPIE*, vol. 11722, pp. 24–40, 2021.
[56] National Weather Service, "Hurricane harvey," 2017. Accessed: Aug. 14, 2024. [Online]. Available: https://www.weather.gov/hgx/hurricaneharvey
[57] ArcGIS, "Hurricane harvey disaster map," 2017. Accessed: Aug. 14, 2024. [Online]. Available: https://www.arcgis.com/apps/View/index.html?appid=8350c2f309bb49f8865a44cb972024c2
[58] Esri, "Houston population density," 2017. Accessed: Oct. 11, 2024. [Online]. Available: https://www.arcgis.com/apps/mapviewer/index.html?webmap=85a821d13a4f4502a85f71c4aae8bae8
[59] A. I. Ameur, O. S. Oubbati, A. Lakas, A. Rachedi, and M. B. Yagoubi, "Efficient vehicular data sharing using aerial P2P backbone," *IEEE Trans. Intell. Veh.*, vol. 10, no. 1, pp. 413–426, Jan. 2025.
[60] K. Messaoudi, A. Baz, O. S. Oubbati, A. Rachedi, T. Bendouma, and M. Atiquzzaman, "UGV charging stations for UAV-assisted AoI-aware data collection," *IEEE Trans. Cogn. Commun. Netw.*, vol. 10, no. 6, pp. 2325–2343, Dec. 2024.
[61] C. Dutriez, O. S. Oubbati, C. Gueguen, and A. Rachedi, "Energy efficiency relaying election mechanism for 5G Internet of Things: A deep reinforcement learning technique," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2024, pp. 1–6.



Md Safwan Mondal received the B.E. degree from Jadavpur University, Kolkata, India, in 2021. He is currently working toward the Ph.D. degree in mechanical and industrial engineering with the University of Illinois Chicago, Chicago, IL, USA. He was an Intern with IIT Mandi, Mandi, India. His research interests include robotics, multi-robot systems, optimization, and AI. He has authored or coauthored in conferences such as IROS, ICRA, RSS and journals including JIRS, with emphasis on multi-robot planning and optimization.



Subramanian Ramasamy received the B.Eng. degree from the Sri Sivasubramaniya Nadar College of Engineering, Anna University, Chennai, India, in 2019, and the Ph.D. degree in mechanical and industrial engineering from the University of Illinois Chicago, Chicago, IL, USA. He was an Operations Research Intern with Amtrak in 2022. His currently with Alstom Transportation as a Fleet Support Engineer. He has authored or coauthored in leading robotics conferences and journals, including IROS, CASE, ICUAS, and JIRS. His research interests include operations research, deep learning, optimization, and autonomous vehicle path planning.



James D. Humann received the B.S. degree in mechanical engineering from The University of Oklahoma, Norman, OK, USA, and the M.S. and Ph.D. degrees in mechanical engineering from the University of Southern California, Los Angeles, CA, USA. Since 2017, he has been with DEVCOM Army Research Laboratory, where he is currently a Mechanical Engineer with Vehicle Applied Research Branch. His research interests include design, path planning, and simulation for multi-robot systems.



James M. Dotterweich received the B.S. degree in aerospace engineering from the University of Colorado Boulder, Boulder, CO, USA in 2008, and the M.S. degree in mechanical engineering with a focus on robotics from The University of Utah, Salt Lake City, UT, USA, in 2015. He is currently a Robotics Research Engineer with DEVCOM Army Research Laboratory, Aberdeen, MD, USA. His research interests include robotic systems and software development for unmanned aerial and ground vehicles.



Pranav A. Bhounsule received the B.E. degree from Goa University, Taleigão, Goa, in 2004, the M.Tech. degree from IIT Madras, Chennai, India, in 2006, and the Ph.D. degree from Cornell University, Ithaca, NY, USA, in 2012. He is currently an Associate Professor of mechanical and industrial engineering with the University of Illinois Chicago, Chicago, IL, USA. He has authored around 75 peer-reviewed publications. His research interests include estimation, learning and control of robotic systems.